

Template for Computer Programming I Practical Recording

Date : 23/01/2023

Problem Specification : Calculate the interest for the month, the total new balance (previous balance + additional charges + interest), and the minimum payment due of the credit cards.

Assumption : The user inputs are valid.

Limitation : N/A

Input : Input the previous balance and additional charges of the credit card.

Processing : the program read the previous balance, additional charges. then Calculate the interest of the previous total balance and find the minimum payment due of the credit cards.

Output : print the previous balance, additional charge, interest, minimum payment due of the credit cards.

Algorithm

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Step1 :Prompt "Enter the previous balance"
Step2: Initialize variable previousBalance for previous balance
Step3: Read the previousBalance given by user
Step4: Assign input value for previousBalance
Step5:Prompt "Enter the additional charges"
Step6: Initialize variable additionalCharges for additional Charges.
Step7: Read the input additionalCharges given by user
Step8: Assign input value for additionalCharges
Step9:Print "CS CARD International Statement "
Step10:Print "=====");

Step11: Print previous balance as "Previousbalance is:$"
Step12: Print additionalCharges as "Additional Charges: $"
Step13: Initialize variable :
        previousTotalBalance ← previousBalance+additionalCharges;

Step14: if(previousTotalBalance >0) is true
        Initialize variable : interest ← previousTotalBalance*2/100;
Step15 : Print interest as "Interest:$"
Step16: Initialize variable : newBalance ← previousTotalBalance+interest;
Step17: Print as newBalance as "New balance:$"-

Step18:if (newBalance <+50) is True
        Print newBalance as "Minimum payment:$ "

        else if(newBalance >50 && newBalance <300) is True
        Print as "Minimum payment: $50"
    else
```

Initialize variable : newPayment $\leftarrow$ newBalance*20/100;
Print newPayment as "Minimum payment: \$"

Step19: if (newBalance <+50) is False

Print "the previous balance was 0 "

Step20: End the program

Program listing : Programe file **attached**

Test data and expected output : **1. Test data:** Enter the previous balance: 15000
Enter the additional charges: 2000

Expected output: Previous balance is: \$15000.0
Additional Charges: \$2000.0
Interest: \$340.0
New balance: \$17340.0
Minimum payment: \$3468.0

Output obtained for test data : **1. Test data:** Enter the previous balance: 15000
Enter the additional charges: 2000

Obtained output: Previous balance is: \$15000.0
Additional Charges: \$2000.0
Interest: \$340.0
New balance: \$17340.0
Minimum payment: \$3468.0

Discussion : The implementation was successful in meeting the requirements

Conclusion : The program accurately calculates the Interest, New balance and Minimum payment.